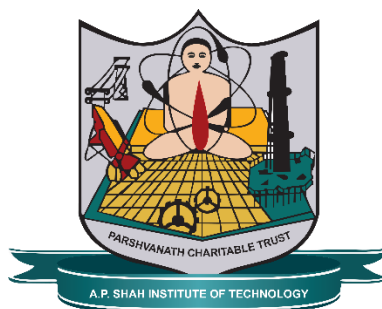


A Project report on

Image Colorization Using Generative Adversarial Networks (GANs)

by

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ABSTRACT

Image colorization is an important task in the field of computer vision and image processing that involves predicting plausible color versions of grayscale images. Traditional methods often relied on heuristic rules, user hints, or example-based transfer, which required extensive human intervention and lacked generalization ability. With the rise of deep learning, end-to-end neural models, especially Generative Adversarial Networks (GANs), have demonstrated impressive capabilities in producing realistic colorized outputs by learning semantic and contextual information directly from data.

In this project, a GAN model is designed and implemented to perform automatic image colorization using a U-Net-inspired generator and a convolutional discriminator. The model is trained on a publicly available dataset of RGB images, where each color image is paired with its grayscale version. The generator attempts to produce realistic color images, while the discriminator learns to distinguish between real and generated outputs. Over time, the adversarial training improves the generator's ability to generate high-quality colorizations. The results show that GANs can successfully capture global color context and generate natural-looking, visually appealing images, demonstrating the potential of deep generative models for color restoration and creative visual enhancement tasks.

INTRODUCTION

Colorization of grayscale images has long been a complex challenge in computer vision because of the inherent ambiguity in mapping luminance values to plausible colors. A single grayscale intensity can correspond to multiple valid color combinations, making the problem highly under-constrained. Early approaches, such as those based on user-provided color scribbles, texture propagation, or exemplar-based transfer, attempted to add colors by referencing similar regions in a set of example images. While these methods achieved reasonable local accuracy, they required substantial manual input and often failed to maintain global color consistency, making them unsuitable for large-scale automation. With the rise of deep learning, particularly Convolutional Neural Networks (CNNs), researchers began formulating colorization as a supervised regression problem, learning direct mappings between grayscale and RGB domains. These networks could automatically infer color patterns based on contextual cues like object shapes, textures, and scene composition. However, CNN-based regression models often produced desaturated or overly smooth results, as their optimization relied heavily on pixel-wise loss functions such as Mean Squared Error (MSE). MSE encourages the network to average over all possible color predictions, penalizing deviations from the ground truth and ultimately leading to washed-out colors.

This project leverages the strengths of GANs to achieve automatic and realistic image colorization. By combining pixel-level reconstruction losses (to ensure structural accuracy) with adversarial feedback (to enforce realism), the system effectively learns both low-level spatial textures and high-level semantic correlations between objects and their expected colors. The resulting colorized images exhibit smoother gradients, more vibrant tones, and a strong resemblance to real photographs, validating the effectiveness of adversarial training in enhancing both aesthetic quality and contextual understanding in colorization models.

Dataset Link

Dataset used: [Image Colorization Dataset – Kaggle](#)

This dataset contains thousands of high-quality RGB images depicting natural scenes, architecture, objects, and other everyday contexts suitable for learning diverse color distributions.

Dataset Description

The dataset is preprocessed by converting each RGB image into grayscale, creating paired data in the form of (*grayscale input*, *RGB target*). In this study, a subset of 2,500 images was used due to hardware limitations in Colab, with each image resized to (120×120)pixels.

Normalization is applied to scale pixel values between 0 and 1, improving gradient stability during training. The dataset is split into 90% training and 10% testing sets. Each grayscale image has a single channel (120,120,1), while the target RGB image has three channels (120,120,3).

Such paired datasets enable supervised learning of colorization, where the model learns to reconstruct missing chrominance information from luminance values.

Legal and Ethical Considerations

The dataset is distributed for public educational use through Kaggle. All images are generic and free of personally identifiable content. The data is used only for academic purposes under the dataset's license terms. In this context, “legal” and “illegal” samples refer not to classifications in the model, but to the ethical use of the dataset. No images depicting sensitive or restricted content are used. The project adheres to data ethics by ensuring non-commercial use, privacy safety, and citation of the dataset source.

Technology Stack:

Programming Language: Python

Deep Learning Framework: TensorFlow–Keras

Numerical Computation: NumPy

Image Processing: Pillow (PIL)

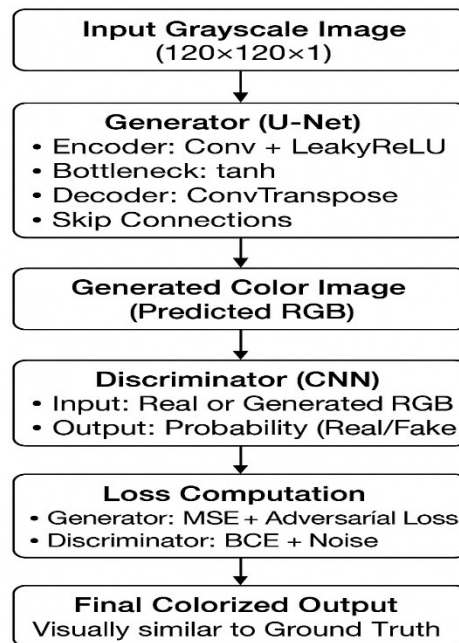
Data Preprocessing: scikit-learn

Development Environment: Google Colab

Hardware Acceleration: GPU (via Colab)

Storage & File Management: Google Drive

Model Architecture:



1. Generator Network-

The **Generator** follows a **U-Net architecture**, which is an encoder–decoder structure with skip connections that transfer fine-grained details from early convolutional layers to later upsampling layers.

- **Input:** Grayscale image (120×120×1)
- **Encoder:** Sequential Conv2D layers with LeakyReLU activations extract features at multiple scales.
- **Bottleneck:** Captures compressed semantic representation using tanh activation.
- **Decoder:** Conv2DTranspose layers reconstruct the color image using upsampling.
- **Skip Connections:** Allow gradient flow and preserve spatial detail.
- **Output:** RGB image (120×120×3)

2. Discriminator Network

The **Discriminator** is a binary classifier CNN that takes either a real RGB image or a generated one as input and predicts whether it is real or fake. It uses several Conv2D

layers followed by ReLU activations, MaxPooling, and fully connected layers with sigmoid output.

This adversarial setup enables the generator to learn how to produce realistic colorizations capable of fooling the discriminator.

Training Methodology

The GAN model is trained through **adversarial learning**, where the generator and discriminator optimize opposing objectives:

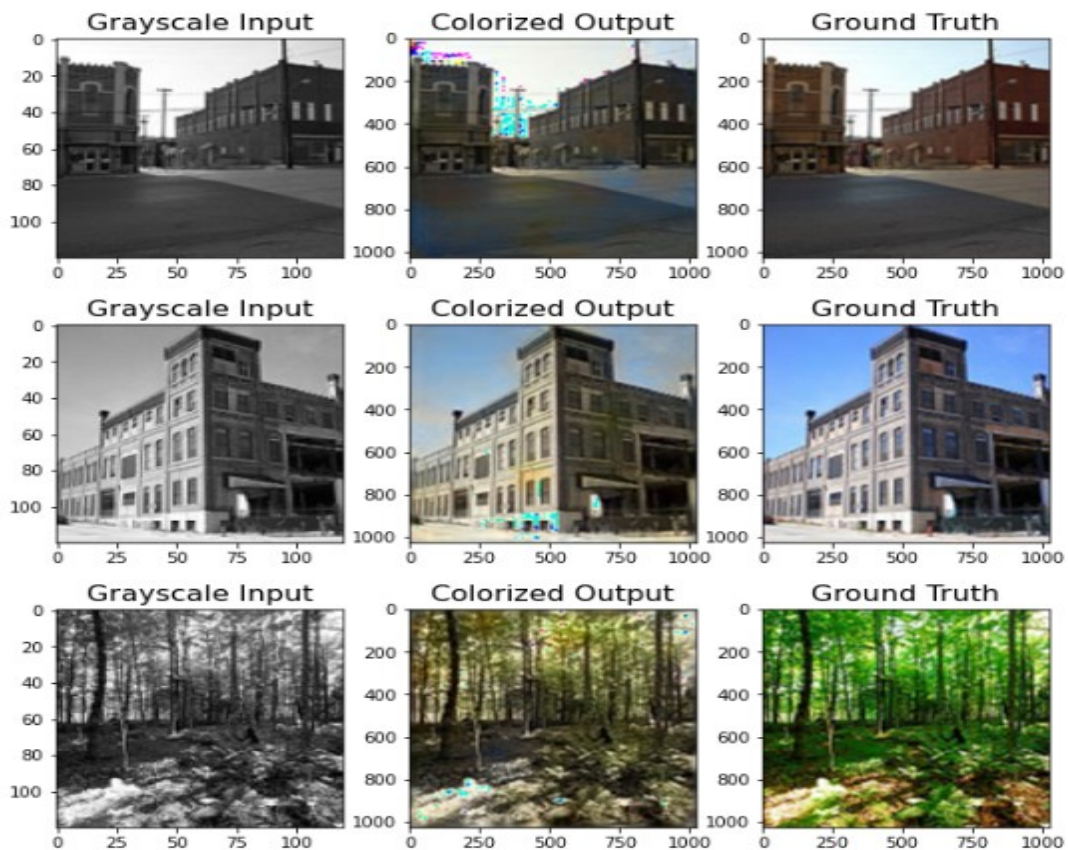
- **Generator:**
 - Minimizes a combination of **Mean Squared Error (MSE)** (to maintain pixel-level fidelity) and **adversarial loss** (to improve realism).
 - Learns both local textures and global semantic information from grayscale input.
- **Discriminator:**
 - Minimizes **Binary Cross-Entropy (BCE) loss** to distinguish real RGB images from generated colorizations.
 - Encourages the generator to produce outputs indistinguishable from real images.

Training parameters:

- Optimizer: **Adam** (learning rate = 0.0005)
- Epochs: **10** (limited by dataset subset and hardware)
- Batch size: **64**
- Stabilization techniques: **Label smoothing** and **Gaussian noise** added to discriminator labels

This adversarial training allows the generator to iteratively improve, producing sharper and more realistic color outputs over time.

OUTPUT



The GAN model demonstrates effective automatic colorization of grayscale images, generating outputs that are visually realistic and semantically consistent. Skies are rendered in shades of blue, vegetation appears green, and objects adopt plausible colors, while fine textures and details are preserved through the U-Net generator's skip connections. Adversarial training enables the generator to produce sharper and more natural colors compared to models trained solely with pixel-wise losses. Some minor limitations, such as slight desaturation or occasional color bleeding, are observed, primarily due to the limited dataset size and the relatively small number of training epochs. Overall, the results confirm that the proposed GAN architecture can successfully capture contextual information, restore plausible colors, and produce high-quality colorized images from grayscale inputs, showcasing the potential of generative models for both creative and practical applications.

CONCLUSION

This project successfully implements a **Generative Adversarial Network (GAN)** for automatic image colorization, using a U-Net generator and a convolutional discriminator. The model effectively learns mappings from grayscale to color images, capturing both low-level textures and high-level semantic context. Adversarial training enables the generator to produce sharp, realistic, and visually appealing colorizations, while the discriminator ensures perceptual consistency.

Experimental results show that skies, vegetation, and objects are colored plausibly, and fine details are preserved. Minor limitations, such as slight desaturation and occasional color bleeding, are observed due to the limited dataset size and number of training epochs. Despite these, the model demonstrates strong potential for restoring grayscale images and enhancing visual content.

Future improvements could include incorporating perceptual losses, training on larger and more diverse datasets, extending the approach to video sequences, and creating real-time applications with user-friendly interfaces. Overall, the project highlights how GANs can transform grayscale visuals into semantically meaningful and aesthetically pleasing images, showcasing the practical and creative potential of deep generative models in image processing.